UNIVERSITY OF DAR ES SALAAM

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES (CoICT)

GROUP WORK

Full-Stack Pipeline for Deploying a Self-Hosted LLM Application

COUSE **Artificial Intelligence**

COUSE CODE **IS365**

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1. Introduction

This report documents the design, implementation, and testing of the University Student Support Assistant, a full-stack application built. The system integrates a local development setup, a locally hosted LLM, a Django REST API backend, a React frontend, authentication, logging, error handling, and automated testing into a single working pipeline.

2. System Use Case

The assistant was built to help university students get quick answers to common administrative and academic questions without needing to search through multiple web pages or contact different offices. The assistant covers eight core areas of student life:

- Course registration procedures
- Examination rules and regulations
- Library services and borrowing policies
- ICT support and account issues
- Hostel application procedures
- Fee payment methods and deadlines
- Academic calendar and important dates
- Student conduct and disciplinary policy

A student interacts with the system through a chat-style web interface, types a natural language question, and receives an AI-generated answer grounded in a system prompt that defines the assistant's role and scope.

3. Tools and Technologies Used

Component	**Technology Used**	**Purpose**
Operating System	Windows	Development environment
Code Editor	Visual Studio Code	Writing and editing all source code
Backend Language	Python 3.10+	Server-side logic
Backend Framework	Django + Django REST Framework	API routing, request handling, authentication
Virtual Environment	Python venv	Isolated dependency management
Local LLM Runtime	Ollama	Serving the language model locally
Model	llama3.2:1b	Generating natural language answers
Frontend Framework	React	User interface and chat experience
Authentication	DRF Token Authentication	Securing the /ask/ endpoint
API Testing	Python requests, Postman	Automated and manual endpoint verification
Version Control	Git and GitHub	Source code management and collaboration

4. System Architecture

The system follows a layered client-server architecture. A student interacts with the React frontend in their browser. The frontend sends an HTTP request containing the student's question to the Django backend. Django authenticates the request using a token, validates the input, and forwards the question (combined with a system prompt) to the locally running Ollama service. Ollama runs the llama3.2:1b model and returns a generated answer. Django logs the interaction, formats the response, and sends it back to the frontend, which displays it in the chat window.

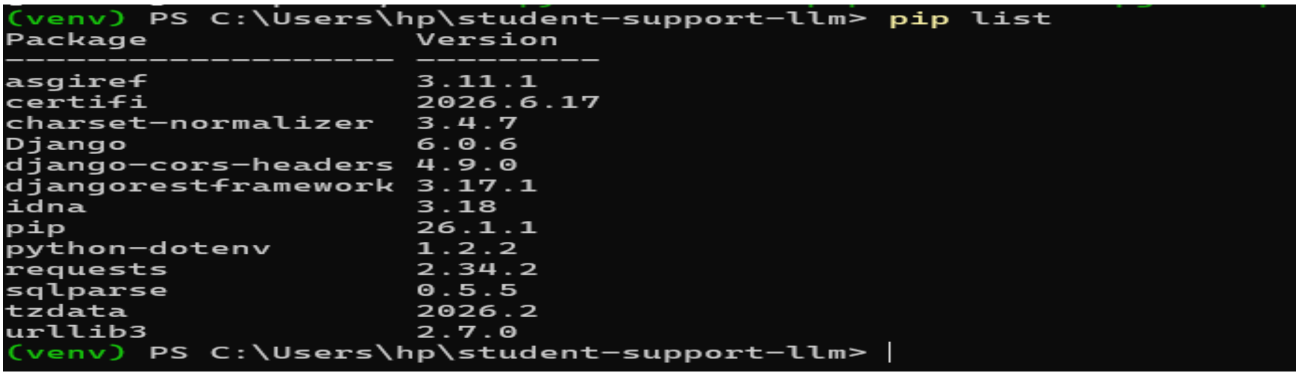
Logical flow: User → Frontend (React) → Backend (Django REST API) → Local LLM (Ollama) → Generated Response → Backend → Frontend → User.

The architecture also includes a configuration layer (Django settings for CORS, authentication, and the Ollama connection details), a logging layer (writing every interaction to a file), and a dedicated error-handling layer that intercepts connection failures, timeouts, and invalid input before they reach the user as raw errors.

5. Implementation Steps

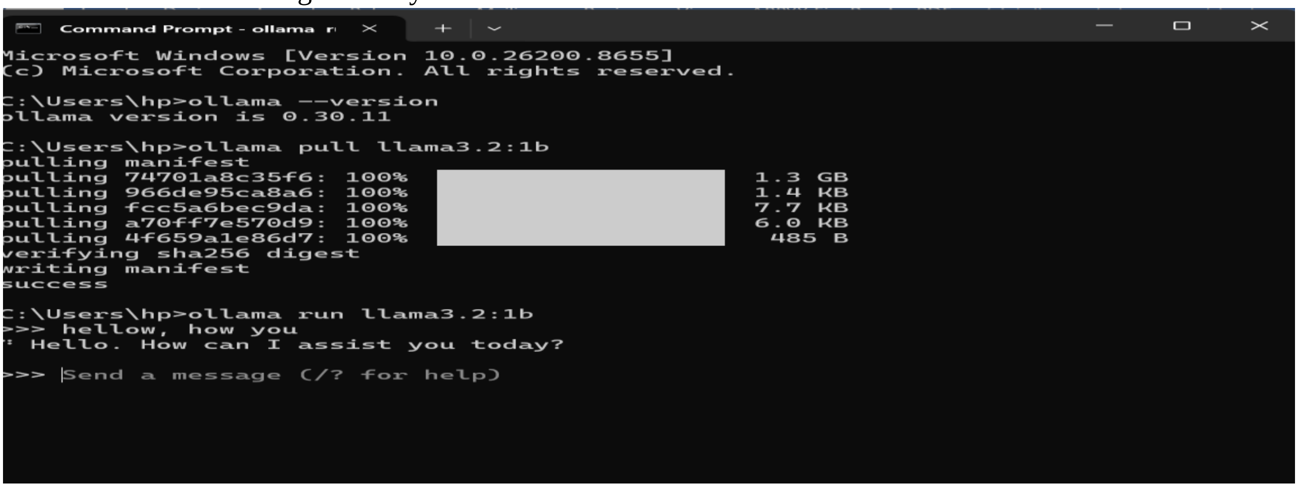
5.1 Environment Setup

A Python virtual environment was created and activated to isolate project dependencies from the system-wide Python installation. Required libraries were installed using pip, including Django, Django REST Framework, django-cors-headers, drf-spectacular, requests, and python-dotenv.



5.2 Local LLM Installation

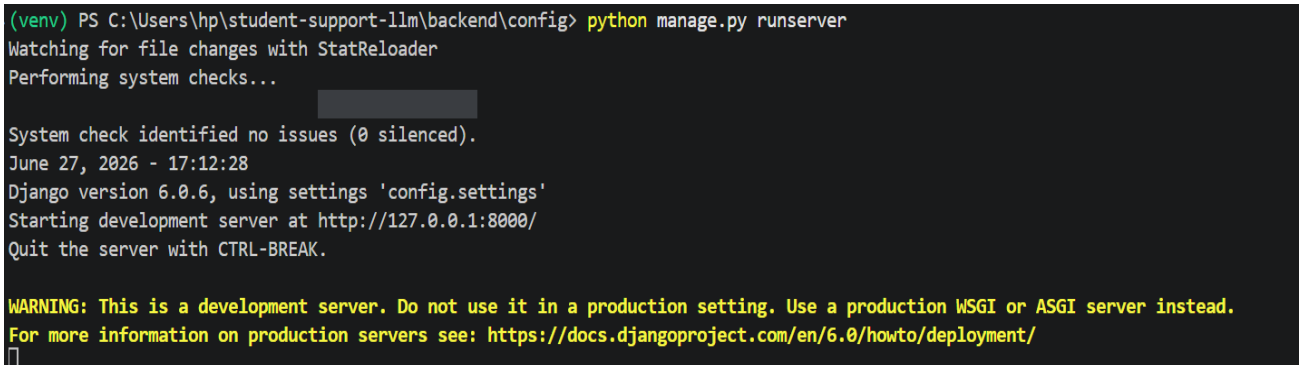
Ollama was installed and the llama3.2:1b model was pulled using the ollama pull command. The model was tested directly through Ollama's built-in API on port 11434 before any backend code was written, to confirm the model was functioning correctly in isolation.



5.3 Django Backend Development

A Django project was created using django-admin startproject, with a dedicated api app handling all business logic. Settings were configured to enable CORS for the React frontend, token-based authentication, structured logging to a file, and the Ollama connection URL and model name. Two main groups of endpoints were implemented: authentication endpoints (register, login, logout, current user) and the core assistant endpoint (ask question), alongside a public health check endpoint.

The llm_client.py module encapsulates all communication with Ollama. It constructs a system prompt that defines the assistant's role and scope, sends the combined prompt to Ollama's generate endpoint, and translates connection or timeout failures into clear, user-friendly exceptions that the view layer can handle gracefully.



5.4 React Frontend Development

The frontend was built using create-react-app and structured into reusable components: an authentication page handling sign up and login, a chat box component managing the conversation state and API calls, and supporting components for message bubbles and loading indicators. The frontend stores the authentication token returned by the backend and attaches it as an Authorization header on every subsequent request to the protected ask endpoint.

5.5 Authentication Layer

To move beyond a prototype-level system, token-based authentication was added using Django REST Framework's built-in Token Authentication. Students register with a first name, last name, email, and password; the backend creates a Django user object and issues a unique token. This token must be included in the Authorization header of every request to the protected /api/ask/ endpoint, ensuring that only logged-in students can use the assistant.

5.6 Connecting the Full Pipeline

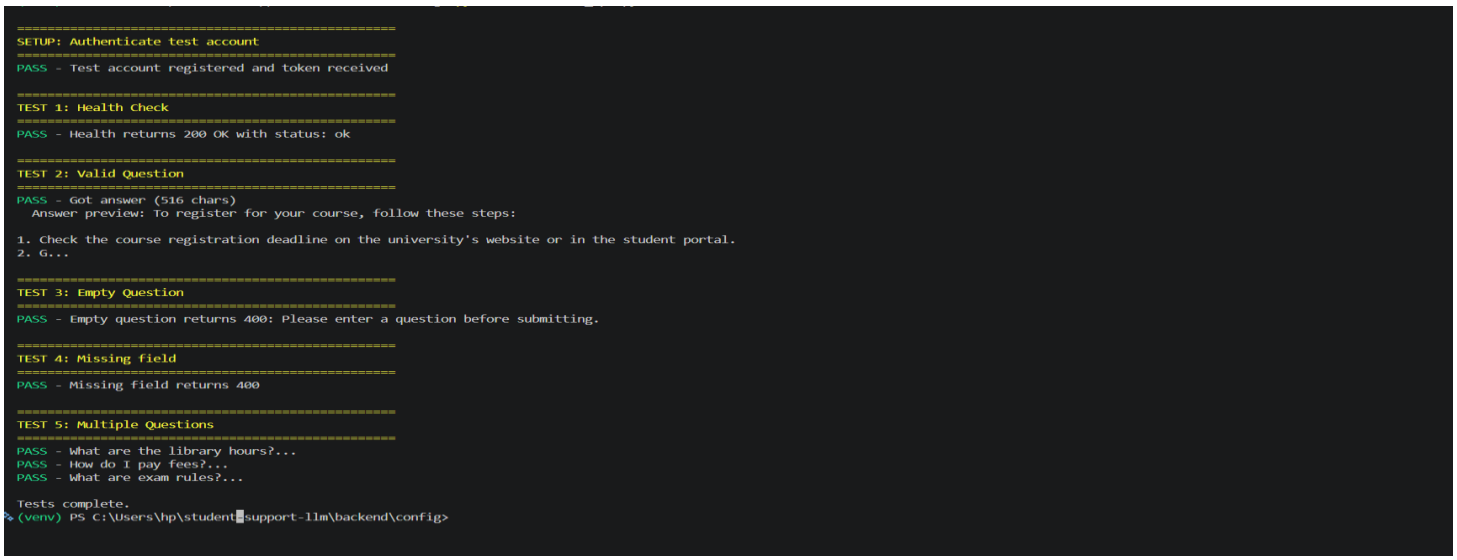
With the backend and frontend running on separate ports (8000 and 3000 respectively), CORS headers were configured to allow cross-origin requests between them. The. env file in the frontend stores the backend API base URL, allowing the connection point to be changed easily without modifying source code.

6. Testing and Results

Three independent testing methods were used to verify the system end-to-end.

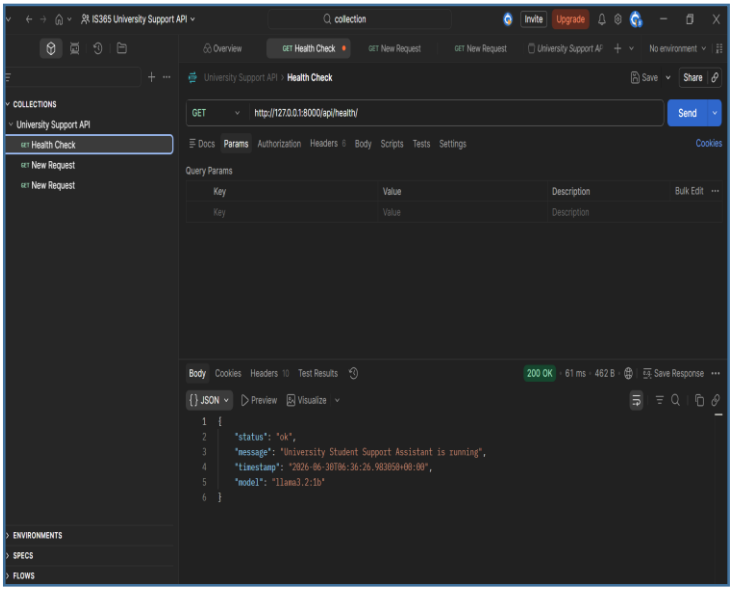
6.1 Automated Python Test Script

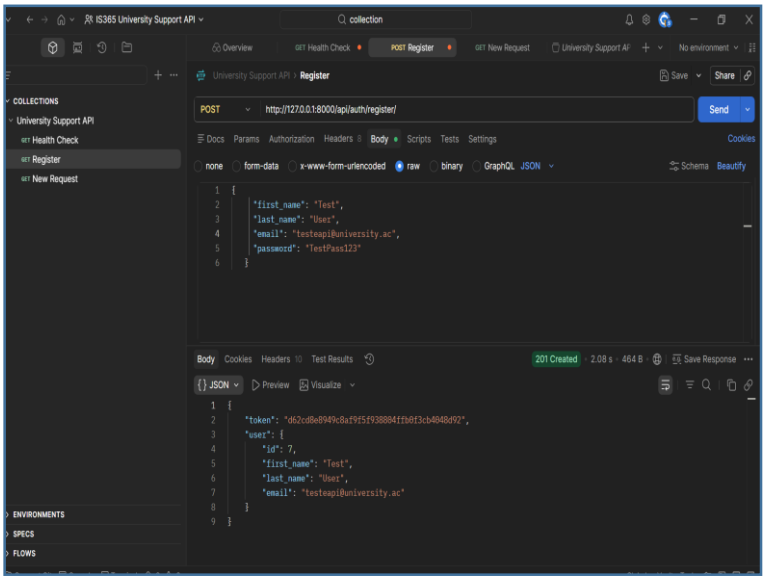
A script located at tests/test_api.py uses the requests library to automatically test five scenarios: the health check endpoint, a valid question returning a meaningful answer, an empty question correctly rejected with a 400 error, a missing field correctly rejected, and multiple sequential questions to confirm system stability under repeated use. The script authenticates first by registering or logging in a test account and attaching the resulting token to subsequent requests.

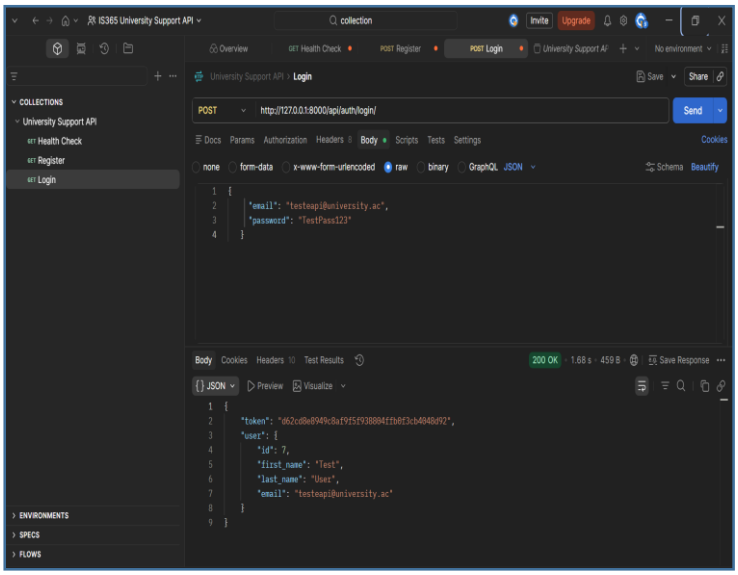


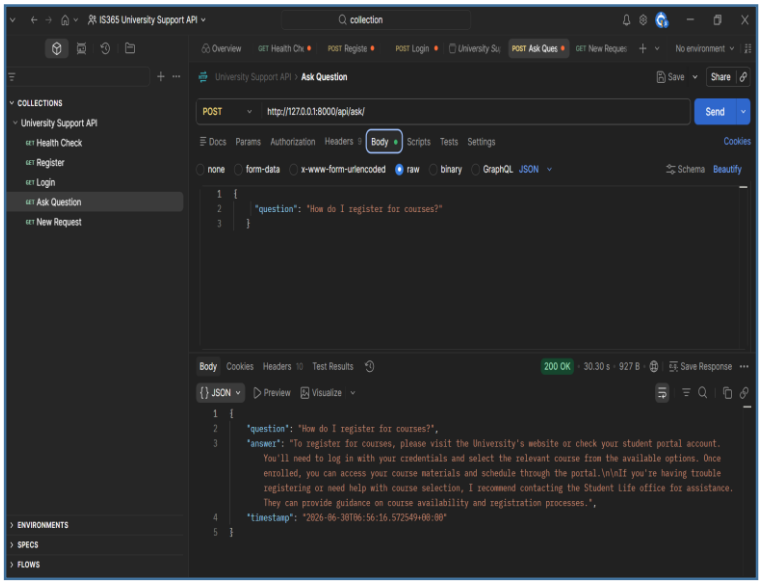
6.2 Postman Collection

A Postman collection was created containing requests for every endpoint, including deliberate failure cases (empty question, missing authentication) to confirm the system responds with appropriate error codes rather than crashing. This collection is included in the project repository for reproducible manual testing.









7. Challenges Encountered

Adding token authentication initially broke the existing automated test script, since the script was not sending an Authorization header; this was fixed by adding a setup step that registers or logs in a test account before running the protected endpoint tests. A Git-related issue also occurred where the frontend folder was mistakenly tracked as a nested Git repository, causing it to appear as a broken link rather than normal files in the GitHub repository; this was resolved by removing the nested. git folder and re-adding the frontend files to the main repository.

8. Production Readiness Discussion

8.1 Main Components of the Deployed System

The deployed system consists of five core components: the React frontend providing the user interface, the Django REST API backend handling routing and business logic, the LLM client module mediating communication with Ollama, the Ollama runtime hosting the llama3.2:1b model, and a logging and error-handling layer recording every interaction and failure.

8.2 Role of Each Component

Django is central to this pipeline because it provides structured URL routing, built-in authentication tools, middleware for cross-origin requests, and a REST framework that converts Python logic into clean JSON API responses. The LLM model is the system's intelligence layer; it interprets the student's natural language question, combined with a system prompt, and produces a relevant response. The frontend is responsible for collecting input, displaying responses in a readable conversational format, and communicating system state such as loading and error conditions to the user.

8.3 Local Hosting Vs External API

Running the model locally through Ollama means no internet connection is required for inference, there is no per-request cost, and student data never leaves the institution's infrastructure, which is significant for privacy. An external API such as OpenAI's would likely produce higher-quality responses and faster inference on capable hardware, but introduces ongoing costs and requires sending student data to a third-party provider.

8.4 Security Risks in an Organizational Deployment

If deployed within a university without further hardening, several risks exist: log files currently store question text in plain, unencrypted form; the Django secret key is stored directly in source code rather than an environment variable; the system runs over unencrypted HTTP rather than HTTPS; and there is no rate limiting, leaving the API vulnerable to being overwhelmed by automated requests.

8.5 Improvements Required Before Production

Before any real deployment, the system would require HTTPS encryption for all traffic, secrets moved to environment variables, a production-grade database such as PostgreSQL replacing SQLite, a production WSGI server such as Gunicorn behind a reverse proxy such as Nginx, request rate limiting, and stricter input sanitisation to prevent prompt injection attacks against the LLM.

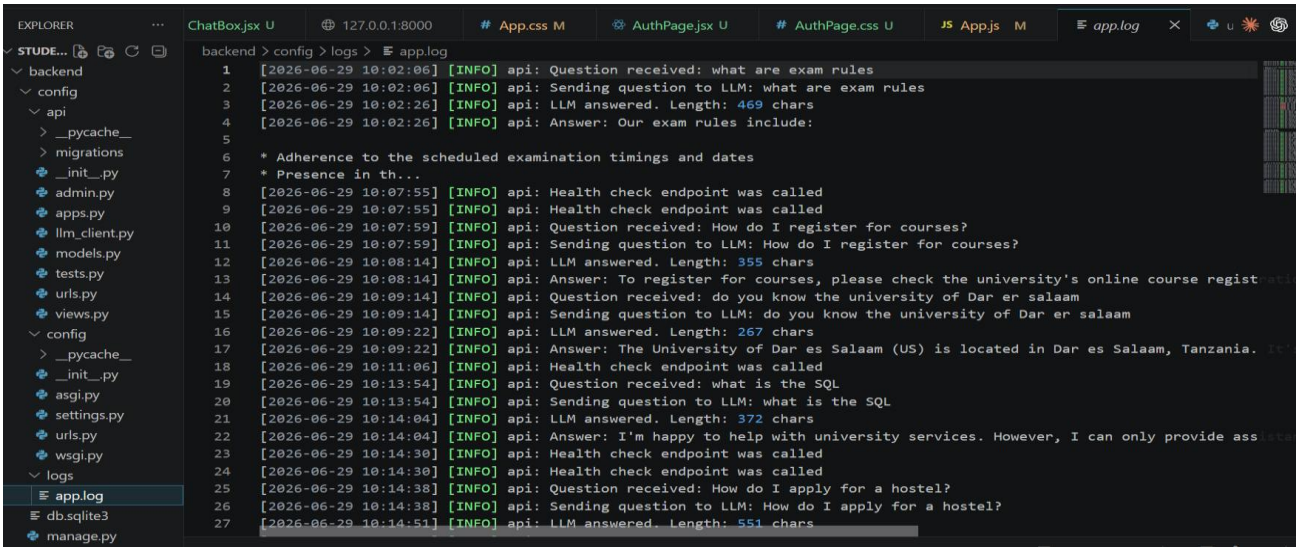
8.6 Monitoring and Data Protection

In real-world use, the system should be monitored using structured log aggregation, response time tracking, and alerting when the Ollama service becomes unreachable. To protect sensitive student information, logs should be encrypted at rest, access to log files restricted to system administrators, and a clear data retention and deletion policy established so that student queries are not stored indefinitely.

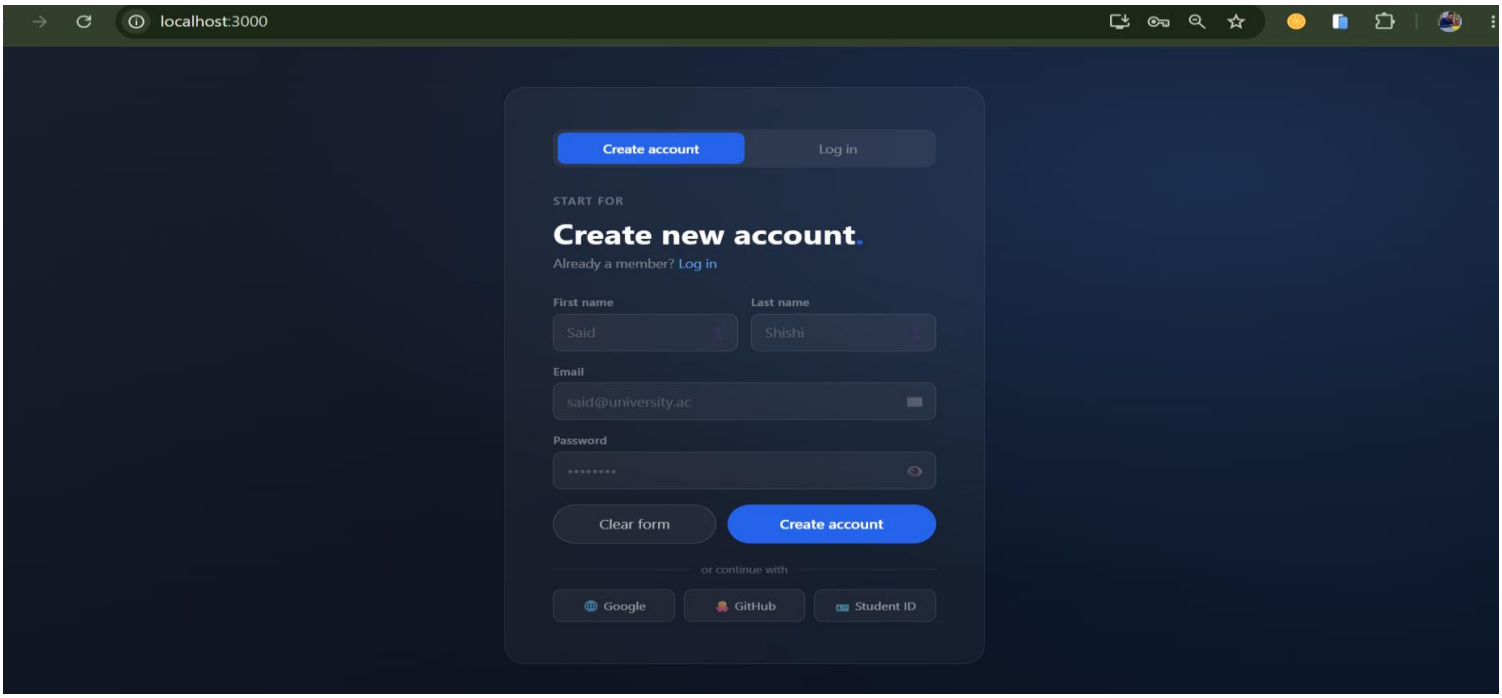
9. Conclusion

This project successfully demonstrates a complete, working pipeline for deploying a self-hosted LLM application, from local model serving through to a secured, tested, and documented web interface. While the assistant itself uses a lightweight model and a relatively simple prompt design, the surrounding infrastructure, authentication, error handling, logging, and the three independent testing methods, reflects the kind of engineering discipline required in production AI systems. The exercise reinforced that building a useful AI application depends far less on the intelligence of the model itself and far more on the reliability of the system built around it.

Appendix G — Logging



Appendix D — Frontend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Create account
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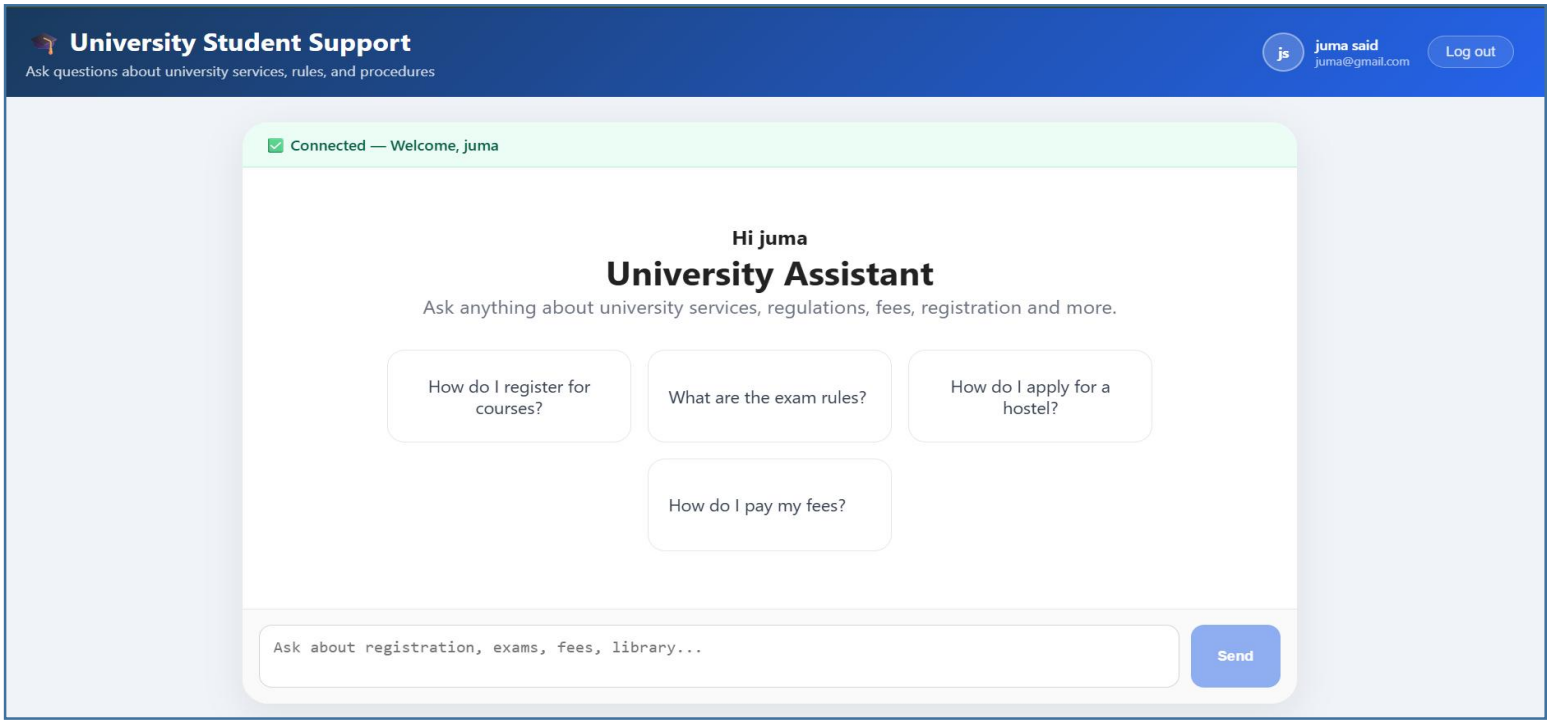
University Student Support
Ask questions about university services, rules, and procedures
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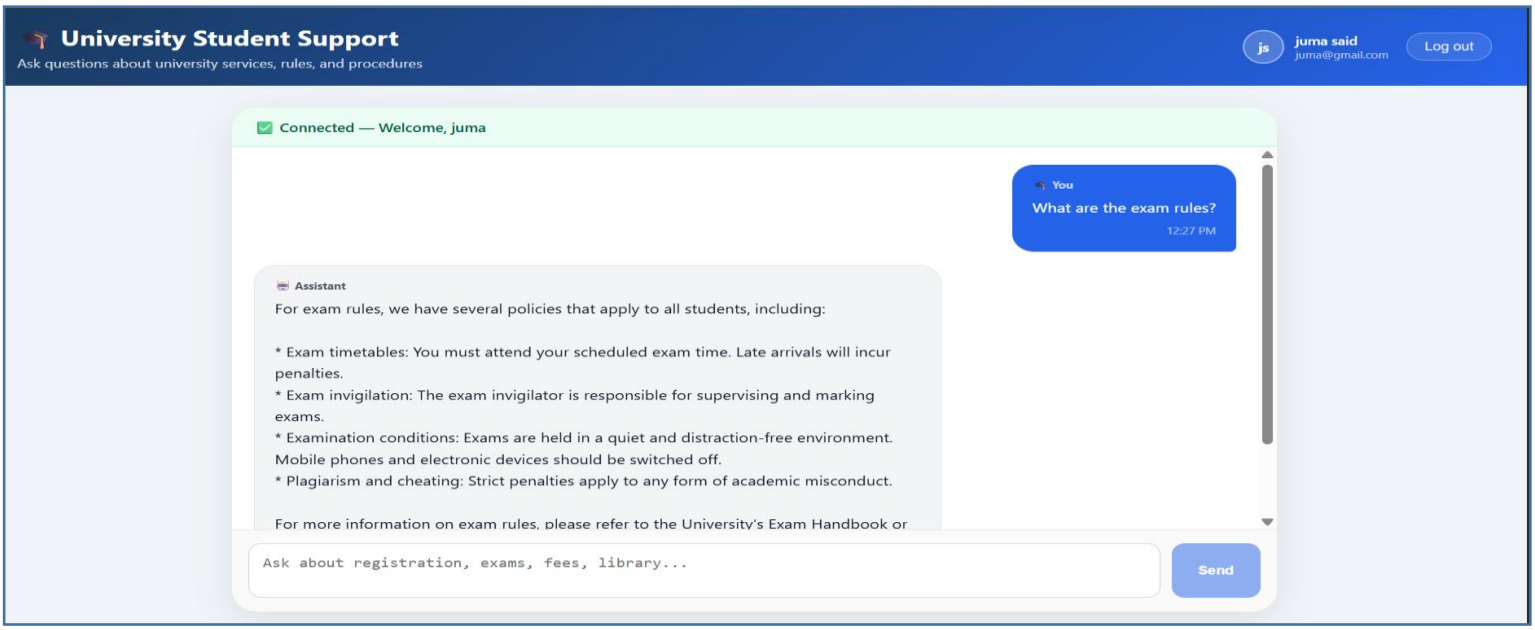
Log out
Connected — Welcome, juma
Hi juma
University Assistant
Ask anything about university services, regulations, fees, registration and more.
How do I register for courses?
What are the exam rules?
How do I apply for a hostel?
How do I pay my fees?
Ask about registration, exams, fees, library...
Send


University Student Support
Ask questions about university services, rules, and procedures
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What are the exam rules?
12:27 PM
Assistant
For exam rules, we have several policies that apply to all students, including:
* Exam timetables: You must attend your scheduled exam time. Late arrivals will incur penalties.
* Exam invigilation: The exam invigilator is responsible for supervising and marking exams.
* Examination conditions: Exams are held in a quiet and distraction-free environment. Mobile phones and electronic devices should be switched off.
* Plagiarism and cheating: Strict penalties apply to any form of academic misconduct.
For more information on exam rules, please refer to the University's Exam Handbook or
Ask about registration, exams, fees, library...
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